PDF Intelligence HuB - AI

Project Setup and Execution Guide

Project Overview

PDF Intelligence HuB - AI is a production-ready, modular AI document analysis application built using Python, Streamlit, and LangChain. It processes user-uploaded PDF files, extracts and cleanses their content, and embeds them into a local vector database (FAISS) for downstream cognitive features: a Retrieval-Augmented Generation (RAG) chatbot with page citations, and an automated multi-format practice Quiz Generator.

Prerequisites

- **Python:** Python 3.12 or newer installed on your machine.
- **API Key:** Google Gemini API Key (default) or OpenAI API Key.

Setup & Installation Steps

Step 1: Extract or Clone Project

Open a terminal and navigate to the project directory:
cd "PDF-Intelligence-HuB-AI"

Step 2: Create a Virtual Environment

Create a clean environment to isolate project dependencies:
On Windows:
python -m venv venv
On macOS / Linux:
python3 -m venv venv

Step 3: Activate Virtual Environment

Activate the created virtual environment:
On Windows (PowerShell):
venv\Scripts\Activate.ps1
On Windows (Command Prompt):
venv\Scripts\activate.bat
On macOS / Linux:
source venv/bin/activate

Step 4: Install Dependencies

Install the required libraries using pip:
pip install -r requirements.txt

Step 5: Setup Environment Variables

Create a file named **.env** in the project root directory (you can copy **.env.example**) and add your API key:
GEMINI_API_KEY=your_gemini_api_key_here
Note: If you wish to use OpenAI, add OPENAI_API_KEY=your_openai_key and change the provider in config.yaml.

Running the Application

Start the Streamlit application server by running the following command in your terminal:

```
streamlit run app.py
```

Once started, the application will automatically open in your default browser at **http://localhost:8501**.

Running Automated Tests

To verify that the modular components are working correctly, run the test suite with pytest:

```
pytest
```

Important Troubleshooting Note: If you receive a pickle or deserialization warning, this is normal behavior for FAISS local databases. The app explicitly sets allow_dangerous_deserialization=True which is safe since you are generating and loading your own local indices.